

ETT

mm	8th	9th
2.5	<1000 g <28 wk	≤1200 g ≤28 wk*
3.0	1000–2000 g 28–34 wk	1201–2200 g 29–34 wk
3.5	>2000 g >34 wk	>2200 g >34 wk

*2.0 mm optional <800 g (22–25 wk)

GA wk	cm	Measure to:
<23	5.0–5.5	9th upper gum 8th upper lip
23–24	5.5	Check: NTL + 1 cm
25–26	6.0	Tip: T1–T2 (text) T2–T3 (program)
27–29	6.5	Flexion ↓ deeper Extension ↑ higher
30–32	7.0	
33–34	7.5	
35–37	8.0	
38–40	8.5	
41–43	9.0	

ETT

mm	8th	9th
2.5	<1000 g <28 wk	≤1200 g ≤28 wk*
3.0	1000–2000 g 28–34 wk	1201–2200 g 29–34 wk
3.5	>2000 g >34 wk	>2200 g >34 wk

*2.0 mm optional <800 g (22–25 wk)

GA wk	cm	Measure to:
<23	5.0–5.5	9th upper gum 8th upper lip
23–24	5.5	Check: NTL + 1 cm
25–26	6.0	Tip: T1–T2 (text) T2–T3 (program)
27–29	6.5	Flexion ↓ deeper Extension ↑ higher
30–32	7.0	
33–34	7.5	
35–37	8.0	
38–40	8.5	
41–43	9.0	

ETT

mm	8th	9th
2.5	<1000 g <28 wk	≤1200 g ≤28 wk*
3.0	1000–2000 g 28–34 wk	1201–2200 g 29–34 wk
3.5	>2000 g >34 wk	>2200 g >34 wk

*2.0 mm optional <800 g (22–25 wk)

GA wk	cm	Measure to:
<23	5.0–5.5	9th upper gum 8th upper lip
23–24	5.5	Check: NTL + 1 cm
25–26	6.0	Tip: T1–T2 (text) T2–T3 (program)
27–29	6.5	Flexion ↓ deeper Extension ↑ higher
30–32	7.0	
33–34	7.5	
35–37	8.0	
38–40	8.5	
41–43	9.0	

v5, Oct 2026

v5, Oct 2026

v5, Oct 2026

UMBILICAL LINES

UAC high T6–T9, low L3–L4
(3 × kg) + 9

UVC IVC–RA junction
[(3 × kg) + 9] ÷ 2 + 1
Revised: drop the +1

Add stump length. Confirm on film.

Cath (Atlas): UAC 3.5 Fr <1.2 kg, UVC 3.5 Fr <3.5 kg;
otherwise 5 Fr

kg	UAC cm	UVC cm
0.5	10.5	6.3
0.75	11.3	6.6
1	12	7
1.25	12.8	7.4
1.5	13.5	7.8
2	15	8.5
2.5	16.5	9.3
3	18	10
3.5	19.5	10.8
4	21	11.5

UMBILICAL LINES

UAC high T6–T9, low L3–L4
(3 × kg) + 9

UVC IVC–RA junction
[(3 × kg) + 9] ÷ 2 + 1
Revised: drop the +1

Add stump length. Confirm on film.

Cath (Atlas): UAC 3.5 Fr <1.2 kg, UVC 3.5 Fr <3.5 kg;
otherwise 5 Fr

kg	UAC cm	UVC cm
0.5	10.5	6.3
0.75	11.3	6.6
1	12	7
1.25	12.8	7.4
1.5	13.5	7.8
2	15	8.5
2.5	16.5	9.3
3	18	10
3.5	19.5	10.8
4	21	11.5

UMBILICAL LINES

UAC high T6–T9, low L3–L4
(3 × kg) + 9

UVC IVC–RA junction
[(3 × kg) + 9] ÷ 2 + 1
Revised: drop the +1

Add stump length. Confirm on film.

Cath (Atlas): UAC 3.5 Fr <1.2 kg, UVC 3.5 Fr <3.5 kg;
otherwise 5 Fr

kg	UAC cm	UVC cm
0.5	10.5	6.3
0.75	11.3	6.6
1	12	7
1.25	12.8	7.4
1.5	13.5	7.8
2	15	8.5
2.5	16.5	9.3
3	18	10
3.5	19.5	10.8
4	21	11.5

v5, Oct 2026

v5, Oct 2026

v5, Oct 2026

Print double-sided, flip on long edge, Actual size. Cut on corner marks.

RSI + RESUS

RSI order Unit

1	Atropine 0.02 mg/kg
2	Fentanyl 1–4 mcg/kg Give slowly (chest wall rigidity)
3	Rocuronium 0.6–1.2 mg/kg

Doses: MacDonald's Atlas. Confirm with order set.

Confirm ETT

HR ↑, chest rise, equal BS, CO₂ color change, CXR

Epi 0.1 mg/mL

IV/IO 0.2 mL/kg (0.1–0.3), flush 3 mL NS

ETT 0.5–1 mL/kg

Repeat every 3–5 min

Volume

NS 10 mL/kg over 5–10 min

Emergency UVC

2–4 cm below skin, to blood return

v5, Oct 2026

RSI + RESUS

RSI order Unit

1	Atropine 0.02 mg/kg
2	Fentanyl 1–4 mcg/kg Give slowly (chest wall rigidity)
3	Rocuronium 0.6–1.2 mg/kg

Doses: MacDonald's Atlas. Confirm with order set.

Confirm ETT

HR ↑, chest rise, equal BS, CO₂ color change, CXR

Epi 0.1 mg/mL

IV/IO 0.2 mL/kg (0.1–0.3), flush 3 mL NS

ETT 0.5–1 mL/kg

Repeat every 3–5 min

Volume

NS 10 mL/kg over 5–10 min

Emergency UVC

2–4 cm below skin, to blood return

v5, Oct 2026

RSI + RESUS

RSI order Unit

1	Atropine 0.02 mg/kg
2	Fentanyl 1–4 mcg/kg Give slowly (chest wall rigidity)
3	Rocuronium 0.6–1.2 mg/kg

Doses: MacDonald's Atlas. Confirm with order set.

Confirm ETT

HR ↑, chest rise, equal BS, CO₂ color change, CXR

Epi 0.1 mg/mL

IV/IO 0.2 mL/kg (0.1–0.3), flush 3 mL NS

ETT 0.5–1 mL/kg

Repeat every 3–5 min

Volume

NS 10 mL/kg over 5–10 min

Emergency UVC

2–4 cm below skin, to blood return

v5, Oct 2026

PICC + FILM

PICC measure

Arm: site → suprasternal notch → 3rd R ICS
(arm 90°, head toward side)

Leg: site → along vein → xiphoid

Tip targets on film

ETT	T1–T2 / T2–T3
UAC	T6–T9 / L3–L4
UVC	IVC–RA jxn: T9–10 / T8–9
PICC arm	SVC, outside heart
PICC leg	Above L4–5 + lateral

ETT, UVC: textbook / NNP program

9th Start FiO₂

≥35 wk	32–34 wk	<32 wk
21%	21–30%	≥30%

SpO₂ targets

2'	3'	4'	5'	10'
65–70	70–75	75–80	80–85	85–95

v5, Oct 2026

PICC + FILM

PICC measure

Arm: site → suprasternal notch → 3rd R ICS
(arm 90°, head toward side)

Leg: site → along vein → xiphoid

Tip targets on film

ETT	T1–T2 / T2–T3
UAC	T6–T9 / L3–L4
UVC	IVC–RA jxn: T9–10 / T8–9
PICC arm	SVC, outside heart
PICC leg	Above L4–5 + lateral

ETT, UVC: textbook / NNP program

9th Start FiO₂

≥35 wk	32–34 wk	<32 wk
21%	21–30%	≥30%

SpO₂ targets

2'	3'	4'	5'	10'
65–70	70–75	75–80	80–85	85–95

v5, Oct 2026

PICC + FILM

PICC measure

Arm: site → suprasternal notch → 3rd R ICS
(arm 90°, head toward side)

Leg: site → along vein → xiphoid

Tip targets on film

ETT	T1–T2 / T2–T3
UAC	T6–T9 / L3–L4
UVC	IVC–RA jxn: T9–10 / T8–9
PICC arm	SVC, outside heart
PICC leg	Above L4–5 + lateral

ETT, UVC: textbook / NNP program

9th Start FiO₂

≥35 wk	32–34 wk	<32 wk
21%	21–30%	≥30%

SpO₂ targets

2'	3'	4'	5'	10'
65–70	70–75	75–80	80–85	85–95

v5, Oct 2026